

SCIENCE MASTER X

Complete Academic Study Notes • Class 10 Science

CHAPTER 06

Control and Coordination

(Full Chapter)

Nervous System • Reflex Action • Human Brain • Coordination in Plants (Tropic & Nastic Movements) • Hormones in Animals

Target Level: CBSE Class 10

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Chapter Overview & Basic Concepts

All living organisms respond and react to changes in their environment (stimuli).

- Control and coordination in multicellular animals are brought about by two integrated systems: the **Nervous System** and the **Endocrine System**.
- Plants lack a nervous system and muscles; they coordinate using chemical compounds called **Plant Hormones (Phytohormones)**.

NCERT High-Yield Insight: Information from our environment is detected by specialized tips of nerve cells called **Receptors**.

- **Phonoreceptors:** Inner ear (Hearing / Balance)
- **Photoreceptors:** Eye (Light / Sight)
- **Thermoreceptors:** Skin (Heat / Touch / Temperature)
- **Olfactory Receptors:** Nose (Smell)
- **Gustatory Receptors:** Tongue (Taste)

SECTION 1: ANIMALS – NERVOUS SYSTEM & NEURON MECHANICS

The nervous system is composed of a specialized network of nerve cells called **Neurons**, which are the structural and functional units of the nervous system.

1.1 Structure & Working of a Neuron

- **Dendrite:** Acquires information and sets off a chemical reaction that creates an electrical impulse.
- **Cell Body (Cyton):** Processes the impulse and passes it along the axon.
- **Axon:** Longest fiber that transmits the electrical impulse away from the cell body to the nerve ending.
- **Synapse:** Microscopic gap between the axon terminal of one neuron and the dendrite of the next neuron.

Transmission Pathway:

Stimulus → Dendrite → Cell Body → Axon → Axon Terminal → [Chemical Release at Synapse] → Dendrite of Next Neuron

 **Diagram: Neuron / Nerve Cell Structure**

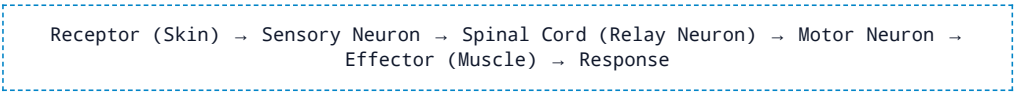


1.2 Reflex Action and Reflex Arc

- **Reflex Action:** An extremely quick, sudden, involuntary, and automatic response to a stimulus in the environment to protect the body from damage.
- **Reflex Arc:** The anatomical pathway taken by nerve impulses in a reflex action.

- **Role of Spinal Cord:** Reflex arcs are formed in the **Spinal Cord** itself for speed, though the information input reaches the brain later.

Path of Reflex Arc:



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SECTION 2: HUMAN BRAIN & CENTRAL NERVOUS SYSTEM

The Central Nervous System (CNS) consists of the **Brain** and **Spinal Cord**, which receive, integrate, and process information from all parts of the body.

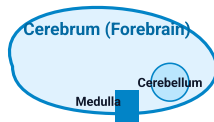
2.1 Divisions and Functions of the Human Brain

Brain Region	Major Sub-parts	Key Functions
Forebrain	Cerebrum	Main thinking part, voluntary actions, memory, intelligence, sensory processing, and center associated with hunger.
Midbrain	—	Controls involuntary actions such as pupillary light reflexes, auditory and visual reflexes.
Hindbrain	Cerebellum	Maintains body posture, equilibrium/balance, and precision of voluntary movements (e.g., riding a bicycle).
Hindbrain	Medulla	Controls involuntary activities like blood pressure, salivation, vomiting, and swallowing.
Hindbrain	Pons	Regulates respiration and acts as a bridge between different parts of the brain.

2.2 Protection of CNS

- **Brain:** Enclosed inside a bony box called the **Cranium (Skull)** and surrounded by fluid-filled layers containing **Cerebrospinal Fluid (CSF)**, which provides shock absorption.
- **Spinal Cord:** Protected inside the bony vertebral column (backbone).

 **Diagram: Human Brain Structure**



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SECTION 3: COORDINATION IN PLANTS (PHYTOHORMONES)

Plants show two different types of movement in response to stimuli:

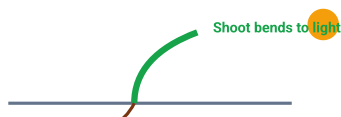
1. **Growth-dependent movements (Tropic Movements)**
2. **Growth-independent movements (Nastic Movements)**

3.1 Tropic vs. Nastic Movements

Feature	Tropic Movements	Nastic Movements
Direction	Directional (depends on direction of stimulus).	Non-directional (independent of stimulus direction).
Growth	Growth-dependent (slow process).	Growth-independent (fast process due to cell turgidity).
Example	Bending of shoot towards light (Phototropism).	Folding up of leaves of <i>Mimosa pudica</i> ("Touch-Me-Not").

3.2 Plant Hormones (Phytohormones) Summary

Plant Hormone	Primary Functions
Auxin	Synthesized at shoot tips; promotes cell elongation . Causes bending towards light.
Gibberellins	Promotes stem growth, leaf expansion, and seed germination.
Cytokinin	Promotes active cell division ; present in high concentration in fruits and seeds.
Abscisic Acid (ABA)	Growth inhibitor ; causes wilting/falling of leaves and induces seed dormancy.



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SECTION 4: HORMONES IN ANIMALS (ENDOCRINE SYSTEM)

4.1 Endocrine Glands, Hormones, and Functions

Gland	Location	Hormone	Principal Functions
Pituitary Gland	Base of Brain	Growth Hormone (GH)	Regulates growth. Deficiency causes Dwarfism ; excess causes Gigantism .
Thyroid Gland	Neck Region	Thyroxin	Regulates metabolism. Requires Iodine . Deficiency causes Goitre .
Adrenal Glands	Top of Kidneys	Adrenaline	"Emergency" hormone. Increases heart rate and blood pressure during stress.
Pancreas	Below Stomach	Insulin	Lowers blood sugar levels. Deficiency causes Diabetes Mellitus .
Testes / Ovaries	Scrotum / Abdomen	Testosterone / Oestrogen	Controls secondary sexual characteristics and reproductive functions.

4.2 Feedback Mechanism

High Blood Sugar Level → Pancreas secretes INSULIN → Glucose absorbed by cells → Blood Sugar drops → Insulin secretion stops (Negative Feedback)

? Important Practice Questions

Q1: What is a synapse? How do electrical impulses travel across it?

Ans: A synapse is a microscopic gap between two neurons. Electrical impulses release chemical neurotransmitters at the axon terminal, which cross the gap to trigger an impulse in the next neuron's dendrite.

Q2: Differentiate between Cerebrum and Cerebellum.

Ans: Cerebrum controls thinking, memory, and voluntary actions. Cerebellum controls physical posture, body balance, and precision of voluntary movements.

Q3: How does Auxin promote bending of a shoot towards light?

Ans: Auxin diffuses towards the shady side of the shoot, stimulating faster cell elongation on that side, which causes the stem to bend towards the light source.

Q4: Why is iodised salt recommended in our diet?

Ans: Iodine is essential for the synthesis of Thyroxin hormone by the thyroid gland, which regulates metabolism. Lack of iodine leads to Goitre.